



- 10 (a) Explain how redshift leads to the idea that the Universe is expanding.

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..... [3]

- (b) Stars in a distant galaxy emit radiation. The total luminosity of the stars in the galaxy is $1.90 \times 10^{36} \text{ W}$.

The emission spectrum of the radiation contains a line X at a wavelength of 658 nm.

Radiation from the galaxy is observed on the Earth. The observed radiation has a radiant flux intensity of $8.42 \times 10^{-16} \text{ W m}^{-2}$. In the observed emission spectrum, line X is at a wavelength of 726 nm.

Determine:

- (i) the distance d of the galaxy from the Earth

$$d = \dots\dots\dots \text{ m [2]}$$

- (ii) the speed v of the galaxy relative to the Earth.

$$v = \dots\dots\dots \text{ ms}^{-1} [2]$$



- (c) Observations of many galaxies, such as the one in (b), lead to many pairs of values of d and v . Plotting these values reveals a trend.

- (i) On Fig. 10.1, sketch the variation of v with d .

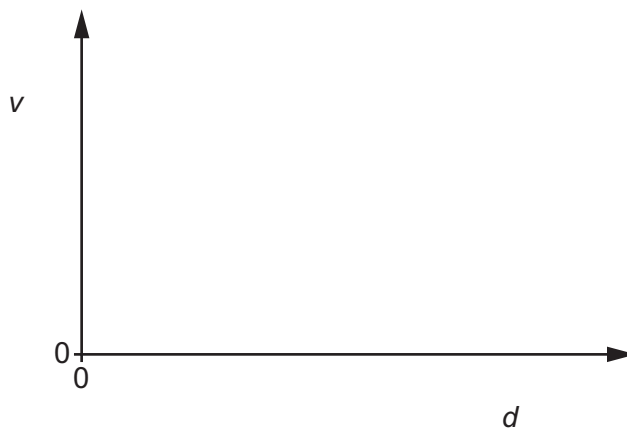


Fig. 10.1

[2]

- (ii) State the name of the quantity represented by the gradient of the line in Fig. 10.1.

..... [1]

[Total: 10]





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